Classical AI Reasoning Agent

AI Assignment 2026 | Roll number M26CSA002
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1. Objective and representation

The agent accepts a natural-language knowledge base and query, and predicts True when its classical engine establishes the requested conclusion. It combines the useful ideas of the two supplied prototypes: indexed closure, context caching, explicit semantics, modular evaluation, and causal reasoning. Their parser and inference defects were repaired in a new implementation.

Input	Representation	Final decision
Facts and rules in English	Signed literals and scoped formulas	Indexed Horn closure; bounded refutation for non-Horn formulas
Optional formal FOL annotations	Formula syntax tree and Skolemized clauses	Own DPLL solver over sound ground instances
Causal statements and interventions	Boolean structural equations and priors	Exact exogenous-state enumeration

Literals keep predicate, ordered arguments and polarity. Variables are distinct from constants. Formulas retain conjunction, disjunction, implication, biconditional, XOR and quantifier scope. Positive and negative facts remain separate. Whole queries are evaluated; no conjunct or rule condition is silently discarded.

2. Classical algorithms

Horn inference. Rules join matching facts by unification; all satisfying bindings are considered. Newly derived facts are indexed and iteration continues to closure. Provenance stores the rule and premises of each conclusion. Repeated queries reuse a bounded context cache. Default OWA requires explicit evidence for a negative statement. Optional CWA uses completed strata and a finite well-founded model for cycles; an undefined negative query is reported as unsupported.

First-order refutation. The engine tests KB AND NOT(query), universally closing free query variables first. Fresh definitional predicates prevent exponential XOR/iff expansion. After NNF, Skolemization and CNF, lifted unit resolution searches for a proof without enumerating all substitutions. Remaining cases use bounded ground instances and our DPLL solver. Equality uses valid congruence axioms; names are not assumed distinct. A refutation proves entailment; a bounded SAT result means no proof was found.

Counterfactual inference. Boolean AND/OR/NOT equations form an acyclic SCM. Exogenous states are enumerated and weighted by Bernoulli priors, filtered by factual observations, then evaluated after replacing intervened equations. Downstream observations are recomputed. The exact posterior is retained; the binary causal convention is probability greater than 0.5.

3. Experiments and measured performance

Development used jise/ruletaker train (1,000 rows), 150 FOLIO training rows, and CounterBench development groups. CounterBench question_id modulo 200 identifies templates across five name substitutions; residues modulo 5 below 3 form development, with the rest held out from that partition. Public evaluation examples were inspected during the audit and improvement pass. These scores are public benchmark measurements after error analysis, not untouched validation or private-test estimates.

Public evaluation	Correct / N	Accuracy	Reasoning (s)	Total (s)
RuleTaker linked test	244/250	97.60%	0.0729	0.4209
RuleTaker linked OOD	373/375	99.47%	0.1125	0.6023
RuleTaker alternative depth-1	200/200	100.00%	0.0050	0.0483
FOLIO, deterministic NL	142/203	69.95%	0.1746	0.2707
FOLIO, formal annotations	170/203	83.74%	6.3845	6.5028
CounterBench alpha, all	950/1000	95.00%	0.0289	0.1038
CounterBench backdoor, all	180/200	90.00%	0.0056	0.0262

FOLIO is scored on all 203 validation rows under binary entailment (Uncertain maps to False). For direct comparison with the original report's 134 True/False rows, formal-annotation accuracy is **76.12% (102/134)**; language-only accuracy is 56.72%. Formal annotations are optional additional input and must not be assumed available on the private test set.

Compared with the preceding release, language-only accuracy increased from 66.01% to 69.95%. Entailment recall rose from 5.56% to 22.22%; precision changed from 80.00% to 76.19%. Broader coverage finds more valid conclusions, with additional false positives. Timings below/above are measured, not a claim of universal speedup.

Alpha held-out groups: 95.00% (380/400).

Backdoor held-out groups: 87.50% (70/80).

Baseline audit: solution_zip scored 38.4% on the linked RuleTaker test, 89% on alpha, 62.5% on backdoor and 50.75% on the FOLIO 134-row formal subset. jalani scored 51.6% on the linked RuleTaker test and 100% on the small alternative depth-1 prefix, but had no dedicated CounterBench or FOLIO implementation. The new CWA engine deliberately flags unsupported negative cycles instead of silently forcing a result.

Runtime: Python 3.14.3, Windows AMD64. Timings are measured in the executed standalone notebook. Every batch begins with a fresh agent, with within-batch context caching enabled. Total includes parsing and evaluation overhead; reasoning time excludes parsing. Download and file-loading time are excluded. These are single-run local timings, not hardware-independent speed claims.

4. Generated reasoning traces

Assignment example. Bob is young and kind; young implies smart; smart and kind imply successful. The repaired parser handles the comma followed by then.

```
successful(bob) <= ((smart(?x) and kind(?x)) implies successful(?x))
  smart(bob) <= (young(?x) implies smart(?x))
    young(bob) <= fact
  kind(bob) <= fact
Prediction: True
```

Quantifier instantiation. Premises: for every x, P(x) implies Q(x); P(alice). Query: Q(alice).

```
Lifted unit resolution refuted KB and negated query (3 unifications, 6 clauses).
Prediction: True
```

Abduction before intervention. B causes A; A OR B causes C; C causes D. Prior P(B)=0.2; C was observed. Under do(A=False), observing C identifies B=True, so D still occurs.

```
Abduction: 1 exogenous assignments consistent with observations {'c': True}.
Action: do({'a': False}); replace intervened equations.
Deduction: recompute endogenous variables in topological order.
P(d=True | observations, intervention) = 1.000000.
Binary causal decision uses probability > 0.5; exact probability is returned separately.
Prediction: True
```

5. Correctness checks and repairs

The final suite contains **122 passing tests**. Independent exhaustive checks cover 150 SAT instances and 80 complete Boolean entailment problems across the two proof methods, including XOR/iff transformation and lifted resolution. Regressions cover the assignment example, negation scope, alternative bindings, universal queries, existential witnesses, equality, causal interventions and cache isolation.

The expanded language parser handles restricted universal subjects, relative clauses, shared subjects and verbs, progressive forms, existential queries, explicit lists and elliptical neither. Categorical statements can use classical refutation when Horn closure is insufficient. Parsed contexts are cached without sharing mutable query state. Labels and reference rationales are tested to have no effect on predictions; all result tables come from actual execution.

6. Limitations and error analysis

Natural-language coverage. FOLIO contains complex English, ellipsis, comparative statements and background assumptions outside the deterministic grammar. Its language-only recall remains limited. The separate public paraphrase evaluation in the new audit still scores near chance; the measured gains primarily concern reasoning over controlled language, unseen knowledge bases and longer rule combinations. Valid syntax from a learned translator also does not guarantee faithful meaning.

Bounded proofs. First-order logic is not generally decidable. This implementation bounds term expansion, ground clauses, equality terms and elapsed reasoning time. Failure to find a proof, a timeout, and unsupported input are recorded separately, although the required binary output maps them to False. It does not claim a complete theorem prover for arbitrary first-order formulas.

Malformed input. Some public FOL annotations have unmatched parentheses, inconsistent predicate spellings or disagreement with their English. Entire unsupported English or formal premises can be omitted with a diagnostic; proof from the retained formulas remains valid for those formulas. The heuristic English translation itself is not guaranteed correct. Causal cycles, incompatible equations and impossible observations are rejected. Unspecified causal root priors use independent Bernoulli(0.5).

Public label disagreements. Some RuleTaker errors have direct derivations despite a False public label. Such examples retain their original labels in reported accuracy. There are no record-specific overrides, name-based answer rules, or private-set predictions. The remaining errors and diagnostics are saved per example for inspection.

7. Reproduction and submission

Run reasoning_agent.ipynb from top to bottom with Python 3.10+. It includes the complete engine, tests, compressed public JSON snapshots and evaluation, so no src folder, network connection or API key is required. It prints final accuracy and inference time and writes submission.csv. Set INPUT_PATH to a new CSV/JSON/JSONL file with context and statement; id is optional. The included CSV contains 250 public RuleTaker demonstration predictions, not private-test results.

The full project additionally contains source code, tests, development/evaluation snapshots, per-example metrics, data checksums, a downloader, a prediction CLI and this reproducible build script. Input and output schemas and semantics settings are documented in README.txt. The notebook was executed in an isolated temporary directory by a fresh Python interpreter and all cells completed successfully.

References and dataset access

Assignment_AI_2026 (2).pdf, supplied course assignment, pp. 1-2.

RuleTaker linked repository: https://huggingface.co/datasets/jise/ruletaker

Additional RuleTaker repository: https://huggingface.co/datasets/tasksource/ruletaker

CounterBench: https://huggingface.co/datasets/CounterBench/CounterBench (question files loaded separately from metadata).

FOLIO: https://huggingface.co/datasets/yale-nlp/FOLIO; public mirror used: https://huggingface.co/datasets/tasksource/folio.

Classical methods: forward chaining and unification; Davis-Putnam-Logemann-Loveland SAT search; Skolemization and ground refutation; structural causal abduction-action-prediction.

8. Broader approaches and generalization study

Neural translation, classical decision. Qwen3-4B-Instruct-2507 (Q4_K_M) receives only English premises and queries. A constrained JSON schema requires one indexed formula per sentence/query. The validator checks coverage, formula syntax, variable binding and consistent predicate arity. It can request up to two repairs with structural error feedback. It never provides gold labels or reference explanations. Our own ground-refutation/DPLL engine computes every final prediction.

Search ablation. A separate first-order resolution implementation adds factoring, standardized-apart variables, equality axioms and age/weight clause selection. On the 203 public formal-input cases it scored 169/203 (83.25%), compared with 170/203 (83.74%) for the existing solver, with longer proof-search time. It remains optional; the selected language pipeline uses the existing ground solver. Search bounds are explicit and no complete first-order decision procedure is claimed.

Matched language-only evaluation	N	Deterministic	Local parser + classical proof
Development: 6 stories	21	14/21 (66.67%)	19/21 (90.48%)
Reserved: 16 stories	48	32/48 (66.67%)	33/48 (68.75%)
Public validation: 73 stories	203	142/203 (69.95%)	162/203 (79.80%)

The 48 reserved queries come from 16 FOLIO training stories outside the earlier 150-row development prefix. Story IDs were selected by a deterministic hash before these experiments; no story overlaps the six development groups. Parser configuration was frozen after development and before reserved scoring. Public validation had been inspected in prior passes and is not untouched. This modest reserved sample supports a limited generalization check, not a private-test guarantee. False and Uncertain both map to False in all rows above.

Public-validation entailment precision: **84.44%**; recall: **52.78%**. Translation failures use the deterministic fallback and are explicitly counted in the saved records. Remaining errors can come from incorrect relations, missing facts, scope, unsupported syntax or proof budgets.

Measured cost. Reserved: translation 89.89s, classical reasoning 0.288s; Validation: translation 463.78s, classical reasoning 2.309s. GPU translation used an RTX 4060 Laptop GPU and llama.cpp b11970 CUDA 12.4. Translation time includes HTTP calls, prompt processing, generation and repairs, with shared prompt caching and queries batched per story. Download/model startup are excluded. Solver comparisons were run in the same experiment. The notebook replays recorded translations offline and labels that replay time separately.

Use on new input. The default notebook remains fully offline. For learned translation, call the included setup/server helpers, then set PARSER_MODE="local" or "adaptive" in the CSV cell. Adaptive mode routes supported closed-world forms to the deterministic engine and complex English through the local translator. The complete source archive provides the --parser local CLI. The approximately 2.5 GB model and runtime are downloaded to the user cache, outside the submission. CPU is supported but substantially slower in these development runs. No API key is required.

Method inspiration: LINC (Olausson et al., EMNLP 2023), https://aclanthology.org/2023.emnlp-main.313/; Logic-LM (Pan et al., Findings of EMNLP 2023), https://aclanthology.org/2023.findings-emnlp.248/. These motivate translation followed by symbolic reasoning; the prover here is our own Python implementation. Model: https://huggingface.co/unsloth/Qwen3-4B-Instruct-2507-GGUF. Runtime: https://github.com/ggml-org/llama.cpp. Pinned hashes and full evidence accompany the source archive.

9. Generalization evaluation and selection

The selection used a fixed validation split of **2,602 queries**: RuleTaker 1,748, FOLIO 294, and CounterBench 560. It has 1,265 True and 1,337 False labels. The final fresh evaluation has **2,938 queries**: official RuleTaker test English and grouped held-out FOLIO training stories, with 1,434 True and 1,504 False labels. CounterBench has no unused public split; its 560 validation questions were all previously examined and are not counted in the fresh final set.

RuleTaker knowledge bases were selected from official dev and test configurations at depths 0, 1, 2, 3, 5 and NatLang. Original human-written text was paired with matching verified public annotations; inconsistent/mismatched source groups were excluded as documented. FOLIO components were grouped by formula skeleton and English five-gram similarity, with previously used stories and their connected neighbors reserved out. All questions within each selected knowledge base/story were kept together. Selection seed, exclusions, dataset checksums and sample counts are saved in the full project.

Metric	Baseline	Improved system
Overall accuracy, fresh final	79.37% (2332/2938)	85.94% (2525/2938)
RuleTaker accuracy	81.23%	88.61%
CounterBench accuracy*	97.32%	97.32%
FOLIO accuracy	64.29%	64.29%
Multi-hop accuracy	69.17%	85.73%
Parsing failure rate	435/2938 (14.81%)	435/2938 (14.81%)
Median response latency	6.96 ms	6.36 ms
95th percentile response latency	15.54 ms	16.85 ms
Total evaluation runtime	5.08 s	4.82 s

*CounterBench is the grouped validation/regression set; the benchmark has no fresh public holdout here and all source questions had been examined previously. Its numbers must not be read as unseen-test generalization. Response latency is measured on each same-context query batch and includes parsing and inference. Total runtime includes loading, evaluation and summary writing; model download/startup are excluded.

The frozen candidate improved fresh-final accuracy by **6.57 percentage points**. A paired bootstrap over 320 independent knowledge-base/story groups gives a 95% interval of [4.34, 8.97] points. The same examples were run through the saved baseline and selected candidate. No private assignment test data was accessed.

10. Generalization groups, diagnostics and remaining errors

Final group	N	Baseline	Improved
Controlled English	2012	90.06%	99.80%
Human-written paraphrases	604	51.82%	51.32%
Multi-hop	785	69.17%	85.73%
Depth 0	1111	88.75%	91.45%
Depth 1	720	82.78%	87.36%
Depth 2	381	75.07%	85.30%
Depth 3	239	63.18%	80.75%
Depth 5	75	65.33%	96.00%
Large knowledge bases	1760	78.12%	87.05%

Independent generated chain diagnostics scored 96/96 for the baseline and 96/96 for the improved system. They cover chain lengths 2, 5, 12 and 20, with up to 100 disconnected facts and four equivalent rule wordings. Their labels were computed by separate finite least-fixed-point closure; these are diagnostic checks, not benchmark rows.

On seven paired CounterBench CPU runs, the causal-model cache preserved every prediction and reduced median end-to-end processing on the same 560 validation questions from 110.8 ms to 87.7 ms. Peak evaluator RSS on the fresh final run was 50.4 MiB; the deterministic run uses no model server.

The main accuracy gain comes from fixes to named-subject recognition, variable/pronoun binding in rule bodies, and CWA handling. The parser now refuses to infer a negative conclusion from an incompletely parsed knowledge base. A well-founded three-valued closure handles cycles: conclusions left undefined are reported unsupported. These changes raised controlled-language accuracy to 99.8% on the final fresh set and multi-hop accuracy by 16.6 points. Human paraphrases remain at 51.3%, FOLIO at 64.3%, and unsupported complex formulations are the largest remaining weakness. The selected system generalizes across new rule combinations and knowledge-base sizes, but linguistic generalization is not established.

Recommended next work: improve compositional parsing on genuinely paraphrased rules, add verified diagnostics for nested quantifier scope and contradictory knowledge bases, then validate on a new externally collected public set. Keep the local language model optional: in the measured validation run it raised FOLIO to 73.8% and overall accuracy to 84.5%, with about 12.9 minutes of measured group-processing time. It did not exceed the selected deterministic candidate overall and adds model memory and translation uncertainty.

Appendix - Complete Public Dataset Evaluation

Run date: 2026-11-03 | Frozen deterministic parser + classical reasoner | All downloaded public rows; no training or parser changes

Coverage: RuleTaker 707,935 (train 480,152 / dev 75,872 / public test 151,911); CounterBench 1,200 (alpha 1,000 / backdoor 200); Yale FOLIO 1,204 (train 1,001 / validation 203). Total: **710,339 rows**.

Official split use. RuleTaker train/dev/test are separate. The downloaded FOLIO release has train/validation only; its public test is unreleased. CounterBench alpha/backdoor are challenge collections, not official train/validation/test splits. This deterministic parser has no learned weights; train rows support parser development/coverage review, and no supervised parameter training is claimed.

Dataset / split	N	Correct	Accuracy	P (True)	R (True)	F1 (True)	TN / FP / FN / TP	Median / p95 ms	Unsupported / limited
RuleTaker train	480,152	359,424	74.86%	100.00%	50.02%	66.68%	238,609 / 0 / 120,728 / 120,815	0.809 / 3.969	0
RuleTaker dev	75,872	56,915	75.01%	100.00%	50.32%	66.95%	37,714 / 0 / 18,957 / 19,201	0.195 / 3.722	0
RuleTaker test	151,911	113,775	74.90%	100.00%	50.10%	66.76%	75,485 / 0 / 38,136 / 38,290	0.178 / 3.593	0
CounterBench alpha	1,000	950	95.00%	96.88%	93.00%	94.90%	485 / 15 / 35 / 465	0.162 / 0.307	10
CounterBench backdoor	200	180	90.00%	94.12%	84.21%	88.89%	100 / 5 / 15 / 80	0.177 / 0.404	5
FOLIO train	1,001	645	64.44%	75.81%	12.11%	20.89%	598 / 15 / 341 / 47	1.469 / 5.310	106
FOLIO validation	203	142	69.95%	76.19%	22.22%	34.41%	126 / 5 / 56 / 16	1.357 / 5.660	5

Interpretation. RuleTaker test accuracy is 74.90% (113,775/151,911), with 100% True precision but 50.10% True recall: the parser is conservative and misses many entailed conclusions. FOLIO validation is 69.95% (142/203), with 22.22% True recall and five unsupported/resource-limited rows. CounterBench alpha/backdoor scores are 95.00%/90.00%; these public examples were examined previously and are regression results.

Evaluation caveats. RuleTaker public test was run once with the parser frozen; prior examination of other public RuleTaker subsets is disclosed. FOLIO validation was previously examined. No private assignment test data was read. FOLIO has True/False/Uncertain labels; for this binary entailment evaluation, Uncertain maps to False, so this is not a three-class score. Training splits are descriptive scores only; no parameters were fitted. Summed per-split processing time was 884.25 seconds (14.7 minutes), excluding process startup and report writing. Peak memory was not recorded; latency depends on the local machine.

The implementation is the authored deterministic NLParser and classical Python inference engine. The parser maps supported English into internal formulas; indexed Horn reasoning, bounded first-order refutation/DPLL, and CounterBench structural causal enumeration produce the final decision. No learned answer classifier or generative model produced these predictions.

Reproducibility artifacts: `experiments/evaluate_full_public_datasets.py`, `results/full_public_datasets_20261103/all_metrics.json`, per-split run/metric JSON, CSV summaries, and row-level prediction JSONL. Source dataset and implementation SHA-256 values are recorded in the run metadata. Run from the project root with `python experiments/evaluate_full_public_datasets.py`.

RuleTaker results by reasoning configuration

All RuleTaker rows are reported; configuration groups overlap the split table only as a stratified breakdown. Accuracy differences show that increased depth is a weakness for this deterministic parser.

Split	Configuration	Rows	Correct	Accuracy	P (True)	R (True)	F1 (True)	TN / FP / FN / TP
Train	NatLang	27,363	20,669	75.54%	100.00%	51.13%	67.66%	13,666 / 0 / 6,694 / 7,003
Train	depth-0	60,557	47,343	78.18%	100.00%	57.63%	73.12%	29,367 / 0 / 13,214 / 17,976
Train	depth-1	61,699	45,994	74.55%	100.00%	49.25%	66.00%	30,752 / 0 / 15,705 / 15,242
Train	depth-2	62,331	46,008	73.81%	100.00%	47.97%	64.84%	30,956 / 0 / 16,323 / 15,052
Train	depth-3	62,613	45,962	73.41%	100.00%	46.92%	63.87%	31,245 / 0 / 16,651 / 14,717
Train	depth-3ext	70,000	51,725	73.89%	100.00%	47.93%	64.80%	34,905 / 0 / 18,275 / 16,820
Train	depth-3ext-NatLang	70,000	52,005	74.29%	100.00%	48.75%	65.55%	34,886 / 0 / 17,995 / 17,119
Train	depth-5	65,589	49,718	75.80%	100.00%	51.55%	68.03%	32,832 / 0 / 15,871 / 16,886
Dev	NatLang	3,899	2,947	75.58%	100.00%	51.25%	67.77%	1,946 / 0 / 952 / 1,001
Dev	depth-0	8,718	6,802	78.02%	100.00%	57.36%	72.90%	4,225 / 0 / 1,916 / 2,577
Dev	depth-1	8,870	6,646	74.93%	100.00%	50.01%	66.68%	4,421 / 0 / 2,224 / 2,225
Dev	depth-2	8,906	6,552	73.57%	100.00%	47.49%	64.40%	4,423 / 0 / 2,354 / 2,129
Dev	depth-3	8,896	6,591	74.09%	100.00%	48.27%	65.11%	4,440 / 0 / 2,305 / 2,151
Dev	depth-3ext	11,587	8,614	74.34%	100.00%	48.86%	65.64%	5,774 / 0 / 2,973 / 2,840
Dev	depth-3ext-NatLang	15,510	11,583	74.68%	100.00%	49.49%	66.21%	7,736 / 0 / 3,927 / 3,847
Dev	depth-5	9,486	7,180	75.69%	100.00%	51.32%	67.83%	4,749 / 0 / 2,306 / 2,431
Test	NatLang	7,793	5,897	75.67%	100.00%	51.36%	67.86%	3,895 / 0 / 1,896 / 2,002
Test	depth-0	17,347	13,542	78.07%	100.00%	57.34%	72.89%	8,428 / 0 / 3,805 / 5,114
Test	depth-1	17,788	13,228	74.36%	100.00%	48.95%	65.73%	8,855 / 0 / 4,560 / 4,373
Test	depth-2	17,741	13,113	73.91%	100.00%	47.98%	64.85%	8,844 / 0 / 4,628 / 4,269
Test	depth-3	17,982	13,297	73.95%	100.00%	48.01%	64.87%	8,971 / 0 / 4,685 / 4,326
Test	depth-3ext	23,222	17,220	74.15%	100.00%	48.60%	65.41%	11,546 / 0 / 6,002 / 5,674
Test	depth-3ext-NatLang	31,017	23,125	74.56%	100.00%	49.33%	66.07%	15,442 / 0 / 7,892 / 7,683
Test	depth-5	19,021	14,353	75.46%	100.00%	50.95%	67.51%	9,504 / 0 / 4,668 / 4,849

Configuration names and split definitions are taken directly from the public RuleTaker Parquet rows. The complete confusion counts, group latency and per-row output are available in the machine-readable results directory.